

LESSON 02 / 第二課

Urban Soils and the Rhizosphere: From Sealed Catchments to Root Carbon Cost

城市土壤與根際：從封蓋集水區到根系碳成本

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Urban trees live in soils that are not agricultural profiles with a city pasted on top. They live in sealed catchments, compacted fills, engineered pits and fragments of residual soil. At catchment scale, impervious cover re-routes water and heat. At pit scale, bulk density, oxygen and available water set the volume a root system can occupy. At the millimetre scale of the rhizosphere, exudates and microbial partners mediate nutrient uptake — and are themselves suppressed by phosphorus surplus, salinity and hypoxia. The finest published constraint is energetic: roots in impeded soil spend more carbon per unit of absorbing length. This review moves down that ladder and treats ‘poor soil’ as several different mechanisms, not as a single insult.

城市樹木所住的土壤，不是上面貼了一座城的農業剖面。它們住在封蓋集水區、壓實填土、工程樹穴與殘餘土碎塊之中。集水區尺度上，不透水覆蓋改道水分與熱。樹穴尺度上，容重、氧與有效水設定根系能佔據的體積。根際毫米尺度上，分泌物與微生物夥伴中介養分吸收——本身又被磷過剩、鹽分與缺氧抑制。最細的已發表約束是能量上的：在受阻土壤中，根系每單位吸收長度花更多碳。本綜述沿此梯子下行，把「土壤差」當成幾種不同機制，而不是單一侮辱。

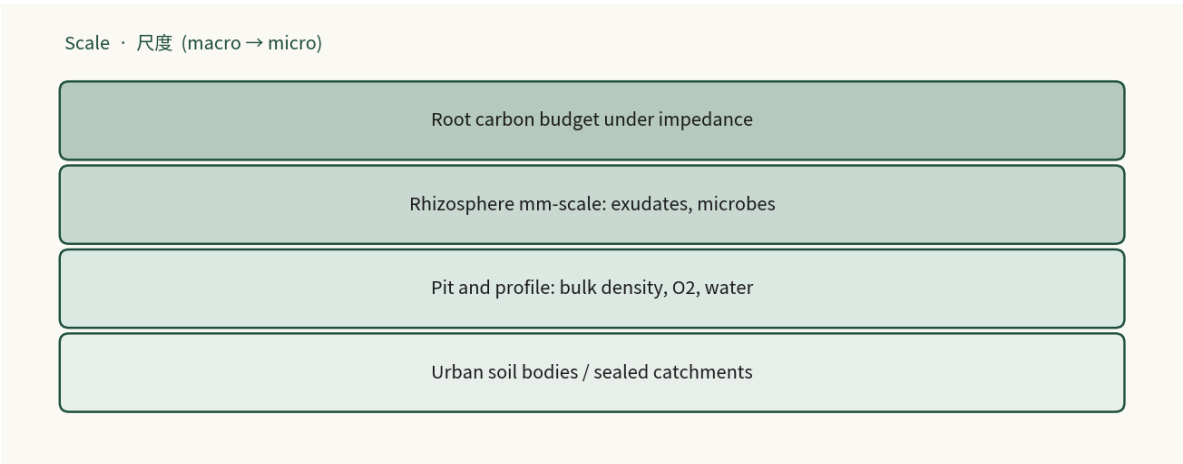


Figure 1. Nested scales from sealed catchments to the root carbon budget.

圖1. 由封蓋集水區到根系碳預算的嵌套尺度。

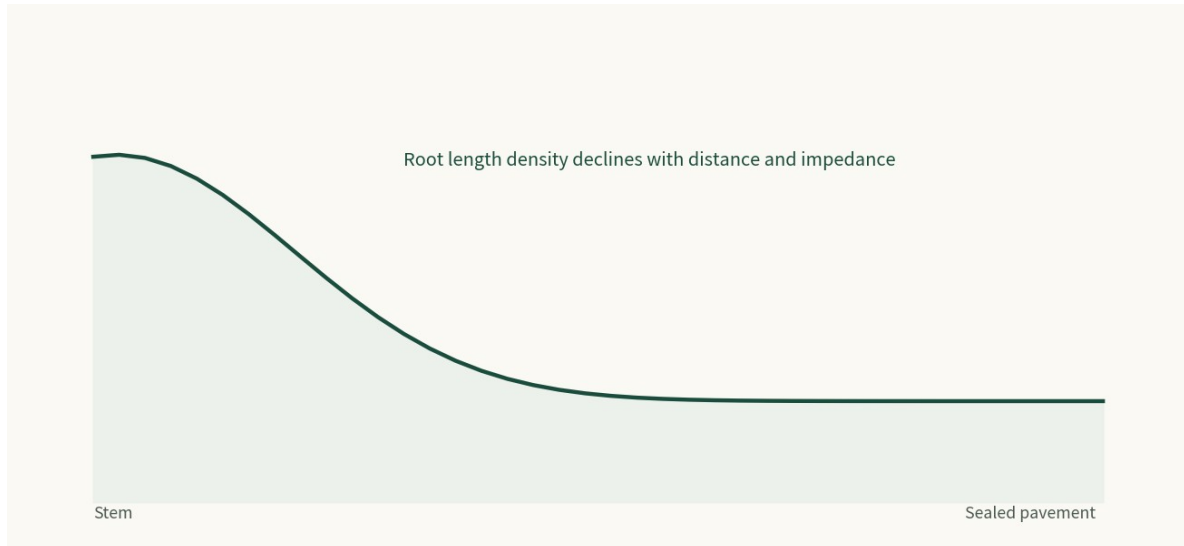


Figure 2. Root occupancy declines with distance from the stem and with mechanical impedance.

圖2. 根系佔用隨離莖距離與機械阻抗下降。

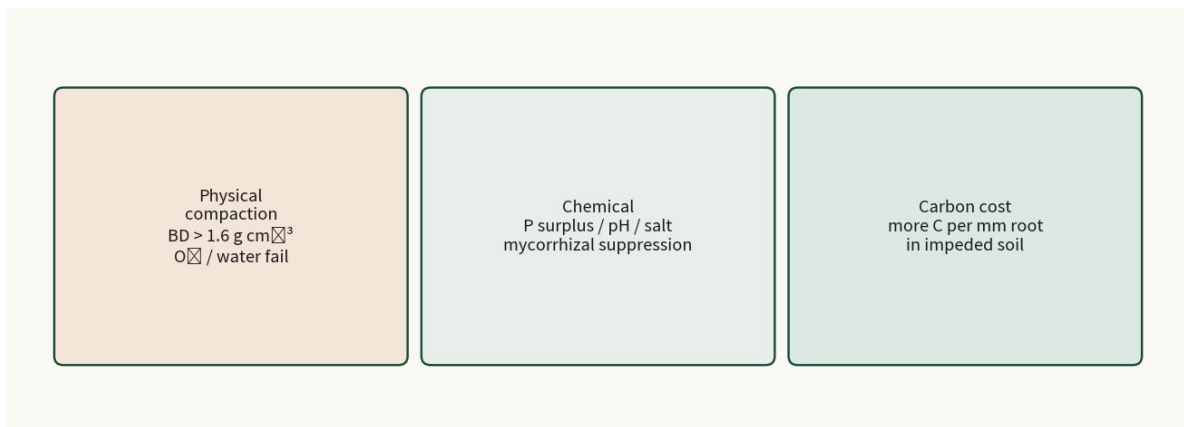


Figure 3. Three non-substitutable soil accounts: physical, chemical, carbon cost.

圖3. 三個不可替代的土壤賬戶：物理、化學、碳成本。

1. Scope

1. 範圍

The object is the soil body that urban roots actually occupy, and the rhizosphere processes measured inside it. Scale runs from sealed catchments to the carbon cost of a millimetre of root (Fig. 1).

對象是城市根系實際佔據的土體，以及在其中測得的根際過程。尺度由封蓋集水區寫到一毫米根的碳成本（圖1）。

2. Catchment and profile

2. 集水區與剖面

Jim's urban-soil programme in humid subtropical cities documented profiles that are anthropogenic from the surface down: mixed construction debris, compacted layers, perched water and sharp texture breaks [1,2]. Craul's earlier synthesis made the same point for temperate cities: bulk density, aeration porosity and available water diverge from agricultural norms even when a texture name is familiar [3]. Sealed surfaces change the water and energy balance of whatever soil remains [4].

Jim 在濕潤亞熱帶城市的城市土壤計劃記錄了從地表向下即為人為的剖面：混合建築碎屑、壓實層、滯水和急劇質地斷裂 [1,2]。Craul 較早的綜述對溫帶城市提出同一點：即使質地名稱熟悉，容重、通氣孔隙與有效水仍偏離農業常態 [3]。封蓋表面改變剩餘土壤的水分與能量平衡 [4]。

The macroscopic result is that an urban ‘soil type’ on a map is not a root map. Two pits of the same mapped unit can differ in impedance and oxygen by enough to change establishment survival. Site description that stops at a texture class has not yet described the object roots use.

宏觀結果是：地圖上的城市「土壤類型」不是根系圖。同一製圖單元的兩個樹穴，阻抗與氧可以差到足以改變定植存活。停在質地級的立地描述，尚未描述根系所使用的對象。

3. Pit scale: density, oxygen, water

3. 樹穴尺度：密度、氧、水

Mechanical impedance and oxygen deficit are coupled. As bulk density rises, air-filled porosity falls; at values commonly reported above about 1.6 g cm^{-3} in medium-textured fills, elongation slows sharply [3,5]. Day, Wiseman and colleagues reviewing urban root growth treated volume restriction and impedance as first-order constraints on the absorbing system, not as aesthetic soil notes [5]. Water can be simultaneously deficient in the sealed matrix and excessive in a perched pit — two stresses that a single ‘irrigate more’ sentence does not distinguish.

機械阻抗與氧虧缺是耦合的。容重上升，充氣孔隙下降；在中質地填土中常報告的約 1.6 g cm^{-3} 以上，伸長急劇減慢 [3,5]。Day、Wiseman 及同事綜述城市根系生長時，把體積限制與阻抗當成吸收系統的一階約束，而不是審美土註 [5]。封蓋基質中可以同時缺水，樹穴滯水中又過濕——單一「多灌溉」句子區分不了這兩種脅迫。

Grabosky and Bassuk’s structural-soil and later suspended-pavement work were attempts to keep load-bearing pavement and a livable root volume in the same plan [6]. They are engineering responses to the pit-scale problem. They are not proofs that any load-bearing mix restores a forest profile.

Grabosky 與 Bassuk 的結構土及後來的懸浮鋪面工作，是試圖在同一平面裡同時保留承載鋪面與可居根體積 [6]。它們是對樹穴尺度問題的工程回應，不是證明任何承載混合料都能恢復森林剖面。

4. Rhizosphere

4. 根際

The rhizosphere is the millimetre-scale volume whose chemistry is modified by exudates, sloughed cells and the microbial community they feed [7]. Phosphorus availability is a control on arbuscular mycorrhizal colonisation: high available P reduces the host’s bid (strigolactones and carbon allocation) rather than ‘killing’ the fungus by contact [8]. Salinity and hypoxia further narrow the microbial partner set. Treating the rhizosphere as a product that can be poured from a bag misses those controls.

根際是分泌物、脫落細胞及其所養微生物群落所修飾的毫米尺度體積 [7]。磷有效性是叢枝菌根定殖的控制因子：高有效磷降低宿主出價（獨腳金內酯與碳分配），而不是靠接觸「殺死」真菌 [8]。鹽分與缺氧進一步收窄微生物夥伴集合。把根際當成可從袋中倒出的產品，會漏掉這些控制。

Field assessment of colonisation (staining and grid-line intersect counts) remains the published method for asking whether a root system is mycorrhizal. A label on an inoculant packet is not that method. The

mycorrhizal account is developed as its own review in Lesson 20; here it is the chemical end of the soil ladder.

定殖的田間評估（染色與格線相交計數）仍是詢問根系是否菌根化的已發表方法。接種劑包裝上的標籤不是該方法。菌根賬戶在第二十課作為獨立綜述展開；此處它是土壤梯子的化學一端。

5. Carbon cost of impeded roots

5. 受阻根系的碳成本

Roots in dense soil construct more cortex and spend more carbon per unit length for less absorbing surface [5,9]. That cost is paid from the same non-structural carbohydrate pool that pays for compartmentalization, flush and defence. A compacted pit is therefore not only a space problem; it is a chronic tax on the carbon account.

密實土壤中的根建造更多皮層，每單位長度花更多碳，換得更少吸收表面 [5,9]。該成本從支付區隔化、抽梢與防禦的同一非結構性碳水化合物庫支付。壓實樹穴因此不只是空間問題，而是對碳賬戶的慢性稅。

The operational implication is ranked interventions: first recover volume, oxygen and a tolerable density; then correct chemistry (P, salt, pH); then, and only then, argue about inoculants. Reversing that order treats a symptom as a cause.

操作含義是有排序的干預：先恢復體積、氧與可耐受密度；再校正化學（磷、鹽、pH）；然後，而且只有在那時，才爭論接種劑。顛倒該順序是把症狀當成原因。

6. Limits

6. 限度

Most published urban-soil numbers are temperate or humid-subtropical. Arid-city fills, tropical lateritic cuts and frozen-ground cities are thinner in the same variables. Transfer of a bulk-density threshold across textures without a water-release curve is a guess, not a measurement.

多數已發表城市土壤數字來自溫帶或濕潤亞熱帶。乾旱城市填土、熱帶紅土切面與凍土城市，在同一變量上較薄。沒有水分釋放曲線就把容重閾值跨質地轉移，是猜測而不是測量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Urban soils in published city profiles are treated here as:

1. 此處把已發表城市剖面中的城市土壤視為：

- A. Agricultural soils with buildings nearby.
A. 附近有建築的農業土。
- B. Anthropogenic bodies with distinct density, water and oxygen regimes.
B. 具有不同密度、水與氧體制的人為土體。
- C. Pure sand by definition.
C. 定義上的純沙。

D. Irrelevant to roots.

D. 與根系無關。

2. Bulk density and oxygen are:

2. 容重與氧是：

A. Independent in compacted fills.

A. 在壓實填土中互相獨立。

B. Coupled: rising density cuts air-filled porosity.

B. 耦合的：密度上升削減充氣孔隙。

C. Equal to P50.

C. 等於 P50。

D. Set by canopy percent.

D. 由樹冠百分比設定。

3. A mapped urban soil unit:

3. 製圖上的城市土壤單元：

A. Is a complete root map.

A. 是完整根系圖。

B. Is not, by itself, a root map.

B. 本身不是根系圖。

C. Equals Wall 4.

C. 等於第 4 壁。

D. Replaces irrigation specification.

D. 取代灌溉規格。

4. High available phosphorus affects AM fungi primarily by:

4. 高有效磷影響叢枝菌根，主要透過：

A. Direct contact poison.

A. 直接接觸毒殺。

B. Reducing the host's signalling and carbon bid.

B. 降低宿主的訊號與碳出價。

C. Raising t/R.

C. 提高 t/R。

D. Increasing flash density.

D. 增加閃擊密度。

5. Roots in impeded soil:

5. 受阻土壤中的根：

A. Spend less carbon per unit length.

A. 每單位長度花更少碳。

B. Spend more carbon per unit absorbing length.

B. 每單位吸收長度花更多碳。

C. Stop using oxygen.

C. 停止使用氧。

D. Become stems.

D. 變成莖。

6. Ranked intervention in this review starts with:

6. 本綜述中有排序的干預始於：

- A. Inoculant packets.
- A. 接種劑包裝。
- B. Volume, oxygen and density.
- B. 體積、氧與密度。
- C. Paint.
- C. 漆。
- D. Lightning conductors.
- D. 避雷導體。

7. Structural soil / suspended pavement is:

7. 結構土／懸浮鋪面是：

- A. Proof that forest profiles are restored.
- A. 森林剖面已恢復的證明。
- B. An engineering response to the pit-scale volume problem.
- B. 對樹穴尺度體積問題的工程回應。
- C. A diversity rule.
- C. 多樣性規則。
- D. A decay fungus.
- D. 一種腐朽菌。

8. Water in urban pits can be:

8. 城市樹穴中的水可以：

- A. Only ever too dry.
- A. 永遠只會太乾。
- B. Simultaneously deficient in the matrix and excessive as perched water.
- B. 在基質中同時缺乏、又以滯水過剩。
- C. Irrelevant.
- C. 無關。
- D. Equal to vapour-pressure deficit.
- D. 等於水汽壓差。

9. Field mycorrhizal assessment uses:

9. 田間菌根評估使用：

- A. Packet labels.
- A. 包裝標籤。
- B. Staining and grid-line intersect counts.
- B. 染色與格線相交計數。
- C. Only i-Tree.
- C. 只有 i-Tree。
- D. t/R.
- D. t/R。

10. Transfer of a bulk-density number across textures without a water-release curve is:

10. 沒有水分釋放曲線就把容重數字跨質地轉移，是：

- A. A measurement.
- A. 一項測量。

B. A guess.

B. 一項猜測。

C. Wall 2.

C. 第 2 壁。

D. A legal register.

D. 法定名錄。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Jim C.Y. urban soil profile series in humid subtropical cities (multiple papers on anthropogenic horizons, compaction and sealed surfaces).
- [2] Jim C.Y. (1998). Urban soil characteristics and limitations in planted trees. Landscape and Urban Planning / related profile papers.
- [3] Craul P.J. (1992). Urban Soil in Landscape Design. Wiley.
- [4] Pouyat R.V., Yesilonis I.D., Nowak D.J. urban soil carbon and cover effects.
- [5] Day S.D., Wiseman P.E., Dickinson S.B., Harris J.R. (2010). Tree root ecology in the urban environment and implications for a sustainable rhizosphere. Arboriculture & Urban Forestry 36: 193–205.
- [6] Grabosky J., Bassuk N. structural soil and later suspended-pavement evaluations.
- [7] Hinsinger P., Bengough A.G., Vetterlein D., Young I.M. (2009). Rhizosphere: biophysics, biogeochemistry and ecological relevance. Plant and Soil 321: 117–152.
- [8] Smith S.E., Read D.J. (2008). Mycorrhizal Symbiosis. 3rd ed. Academic Press. P-supply control on AM.
- [9] Bengough A.G. et al. root elongation versus mechanical impedance and aeration.
- [10] Scharenbroch B.C., Lloyd J.E., Johnson-Maynard J.L. urban soil quality indicators.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。